

The Flipped Interaction Pattern: Letting AI Interview You for Better Database Design

Pharmacy Data Warehouse Design with Claude Opus 4.7

Initial Prompt (using the flipped interaction prompt format):

Help me do a top-down design of my database. Ask me questions one at a time about what I'm trying to do. Based on what I say, decide if it's better to ask me about the types of questions I'm trying to answer with my data or the domain that I'm working in. The goal is to facilitate a great database design. You can ask me one question at a time. Ask the first question.

Claude: What's the main problem you're trying to solve with this database — what will it help you do or decide?

Response: I want to build a database that manages pharmaceutical claims for a health insurance company. The claims data should include GPI and NDC information.

Claude: Are you primarily building this to **process and adjudicate claims in real time**, or to **analyze claims after the fact** (reporting, cost management, fraud detection, etc.) — or both?

Response: Analyze after the fact.

Claude: Who are the main "actors" in your data — for example, are you tracking members (patients), prescribers, pharmacies, and health plans, or is the scope narrower than that?

Response: Members

Claude: Got it — members are the core entity. When you analyze claims for a member, what's the most important question you want to answer? For example: *What drugs is this member taking and what did they cost?* or *Is this member adherent to their medications?* or something else?

Response: Both

Claude: When you look at drug cost for a member, do you need to break it down by **who paid what** — for example, plan paid vs. member copay vs. other payer — or is total cost sufficient?

Response: Break it down

Claude: Do you need to track the member's **plan enrollment over time** — for example, knowing which health plan and benefit design they were on when each claim was filled — or is a simple member profile sufficient?

Response: Yes, that is a requirement

Claude: When you measure **medication adherence**, which approach matters most to you — proportion of days covered (PDC), medication possession ratio (MPR), gap days, or do you need to support multiple adherence methodologies?

Response: PDC

Claude: For PDC calculations, do you need to track adherence at the **drug class or therapeutic category level** (e.g., all statins together) or at the **individual drug level** (specific NDC or GPI), or both?

Response: NDC

Claude: Since you're calculating PDC at the NDC level, do you need to handle **drug switches** within the same therapeutic class — for example, if a member switches from one statin NDC to another, should those claims be grouped together for adherence purposes, or treated independently?

Response: Yes, group them together for adherence purposes

Claude: That means you'll need a way to map NDCs to a therapeutic grouping. Do you want to manage that **drug taxonomy yourself** (maintaining a reference table that maps NDC → GPI → therapeutic class), or will you be **loading it from an external source** like a drug database vendor (e.g., Medispan, First Databank)?

Response: Maintain a reference table myself

Claude: For your reference table, do you already know which **levels of the GPI hierarchy** you want to store — for example, GPI-2 (drug group), GPI-4 (drug class), GPI-6 (drug name), GPI-8 (drug name extended), GPI-10 (dosage form), GPI-14 (full GPI with strength) — or do you want to talk through which levels matter for your use cases?

Response: Full GPI

Claude: Good — full GPI-14 gives you the most granularity. For your NDC reference table, do you need to track **NDC package size and quantity information** (e.g., package size, units per package) in addition to the drug name and GPI mapping, or is the mapping itself sufficient for your analytical needs?

Response: Yes

Claude: When a claim comes in, what **date fields** matter most to you — date the prescription was filled, date it was submitted/processed, date the days supply runs out, or all of the above?

Response: Filled date and days supply

Claude: Good — those two are the foundation of PDC calculation. For the claim itself, what **quantity information** do you need to capture — quantity dispensed, quantity prescribed, or both?

Response: Both

Claude: Do you need to track **refill information** on a claim — for example, refill number or whether it's a new prescription vs. a refill — or is that not important for your analysis?

Response: Yes

Claude: Do you need to capture the **prescriber** on each claim — even if you're not building out a full prescriber entity right now — so you have it available for future analysis?

Response: Yes, prescriber is important

Claude: For the prescriber, what's the minimum you need to capture — just the **NPI** (national provider identifier), or do you also want name, specialty, and/or geographic information?

Response: NPI, specialty, name, plus other demographic information including city and state

Claude: Do you need to track the **pharmacy** where the claim was filled — NPI, name, location — or is the focus purely on the member and prescriber side?

Response: Yes, it is important to track pharmacy information including NPI

Claude: Good. Do you need to distinguish between **pharmacy types** — for example, retail vs. mail order vs. specialty pharmacy — since that can affect days supply logic and adherence calculations?

Response: Yes, that is important

Claude:

We've covered a solid amount of ground. Let me summarize the key entities and relationships I'm hearing before we go deeper into any one area:

Core Entities

- **Member** — with plan enrollment history (effective/term dates, benefit design)
- **Claim** — filled date, days supply, quantity dispensed, quantity prescribed, refill number, new vs. refill flag, cost breakdown (plan paid, member copay, other payer)
- **Drug Reference** — NDC → full GPI-14 mapping, package size, units per package, drug name
- **Prescriber** — NPI, name, specialty, city, state, other demographics
- **Pharmacy** — NPI, name, type (retail/mail/specialty), location

- **PDC Grouping** — a table that groups NDCs into therapeutic cohorts for adherence calculation

Key Relationships

- A member has many claims over time
- A member has enrollment periods (one-to-many)
- A claim links to one drug (NDC), one prescriber, one pharmacy
- NDCs roll up to a GPI-14, which rolls up to a therapeutic group for PDC

Does this accurately reflect what you're building, and is there any entity or relationship here that feels wrong or missing?

Response: The member data should be versioned to track changes to member records, such as marital status, health plan, employer, ETC.

Claude: Good — that's an important distinction. For member versioning, do you want to use a **slowly changing dimension Type 2** approach — where each change creates a new row with effective and expiration dates, preserving the full history — or do you have a different versioning pattern in mind?

Response: Slowly changing dimension Type 2

Claude: For the SCD Type 2 implementation, do you want a **surrogate key** on the member dimension (a system-generated key that uniquely identifies each version) separate from the member's **natural/business key** (like a member ID assigned by the health plan), so that claims can point to the exact version of the member record that was active at fill date?

Response: Yes, a system generated key. A member ID that is constant across member record changes is also important

Claude: That's a clean design — the constant member ID lets you pull all claims for a member across their history, while the surrogate key lets you point to the exact member version active at fill date.

For the member record itself, what **demographic and eligibility attributes** do you want to version — for example, date of birth, gender, marital status, employer, health plan, benefit plan, coverage type (individual/family), zip code? Or do you have a specific list in mind?

Response: Those fields are a good start

Claude: For the **employer** and **health plan** — do you want those as simple attributes on the member record, or do you need them as separate entities with their own tables (e.g., an employer table with employer name, industry, group number, and a health plan table with plan name, plan type like HMO/PPO/HDHP, formulary details)?

Response: Separate tables

Claude: For the **health plan table**, do you need to track **formulary information** — for example, which drugs are covered, at what tier, and what the cost sharing rules are — or is the plan-level detail (name, type, group number) sufficient for your analytical needs?

Response: Yes, it is very important to include the formulary information

Claude: For formulary, do you need to track **tier-level cost sharing rules** (e.g., Tier 1 = \$10 copay, Tier 2 = \$30 copay, Tier 3 = 20% coinsurance) in addition to which NDCs or GPIs are on the formulary, or is just the formulary membership (covered/not covered, tier assignment) sufficient?

Response: Yes

Claude: Formularies change over time — do you need to **version the formulary** as well, so you can look back and know what tier a drug was on and what the cost sharing rules were at the time a claim was filled?

Response: Yes, that also needs to be versioned

Claude: Good — that means formulary will also need effective and expiration dates on both the drug-to-tier assignments and the tier cost sharing rules.

Do you need to track **formulary exceptions or overrides** — for example, when a member gets a non-formulary drug approved through a prior authorization or step therapy exception — so that the claim can be correctly attributed even when it falls outside the standard formulary rules?

Response: Yes, that is very important

Claude: For prior authorization and step therapy exceptions, do you need to track the **full exception lifecycle** — for example, request date, approval/denial status, approving clinician, effective date, expiration date, and the specific drug approved — or just the fact that an exception exists on a claim?

Response: Yes

Claude: We're building out a rich schema. Before we go further, do you need to track **deductible and out-of-pocket accumulator information** at the member level — for example, how much of their deductible and out-of-pocket maximum they've met within a benefit period — since that directly affects the member cost breakdown on each claim?

Response: Yes

Claude: For the accumulator data, do you need to track accumulators **separately by family vs. individual** limits, and also by **benefit period** (typically plan year), so you can accurately reconstruct cost sharing at the time each claim was filled?

Response: Yes

Claude: Do you need to handle **cross-accumulator relationships** — for example, where a member's individual accumulator counts toward the family accumulator — or is tracking them as separate buckets with their own limits sufficient?

Response: Yes, cross-accumulator

Claude: That adds meaningful complexity — you'll need a way to link individual and family accumulator buckets and apply contributions to both when a claim is processed.

At this point we have a fairly comprehensive picture. Let me check one important area we haven't touched — do you need to track **drug utilization review (DUR) or clinical edits** on claims — for example, drug-drug interactions, duplicate therapy, or dosage alerts that were triggered and either overridden or rejected at the point of dispensing?

Response: No

Claude: Good — that keeps the clinical rules engine out of scope. One last major area: do you need to track **claim adjustments and reversals** — for example, when a claim is reprocessed, reversed, or corrected after the original fill — so your cost and adherence calculations always reflect the final adjudicated state?

Response: Yes

Claude: For claim adjustments and reversals, do you want to track the **full adjustment history** — keeping the original claim and each subsequent adjustment/reversal as separate rows with a status and sequence number linking them — or just the **final adjudicated version** of each claim?

Response: Yes

Claude: Good — full adjustment history means every version of a claim is preserved, and your analytics will need to filter to the latest adjudicated version for cost and PDC calculations. That's an important design constraint we'll carry forward.

I think we now have enough to produce a comprehensive top-down database design. Shall I generate the full **entity-relationship design** — entities, attributes, primary/foreign keys, and relationship descriptions — as a structured document?

Response: Yes, please

Claude: Now let me build the comprehensive ERD document.

Ran 3 commands, created a file, read a file

Ran 3 commands, created a file, read a file

Here's your **complete database design document**. It covers all 18 entities across 7 domains, organized as follows:

What's included:

- **Section 1 — Design Overview:** Key architectural decisions (SCD2, claim versioning, formulary versioning, etc.)
- **Section 2 — Entity Inventory:** Quick-reference table of all 18 entities with domain and purpose
- **Section 3 — Entity Definitions:** Full attribute tables for every entity with data types, constraints, and column-level descriptions, organized by domain:
 - Member Domain (MEMBER_DIM, MEMBER_ENROLLMENT, EMPLOYER)
 - Plan Domain (HEALTH_PLAN, FORMULARY, FORMULARY_DRUG, FORMULARY_TIER)
 - Accumulator Domain (ACCUMULATOR, ACCUMULATOR_CONTRIBUTION)
 - Claims Domain (CLAIM, CLAIM_COST, CLAIM_ADJUSTMENT)
 - Drug Reference Domain (DRUG_REFERENCE, PDC_THERAPEUTIC_GROUP, PDC_NDC_GROUP_MAP)
 - Provider Domain (PRESCRIBER, PHARMACY)
 - Authorization Domain (PRIOR_AUTH)
- **Section 4 — Relationships:** Complete relationship table with cardinality and descriptions
- **Section 5 — Analytics Patterns:** Implementation guidance for PDC calculation, point-in-time member lookups, formulary compliance, and accumulator balances
- **Section 6 — Implementation Notes:** Indexing recommendations, SCD2 maintenance procedure, claim adjustment processing, and NDC formatting rules

File: **Pharma claims db design.docx**

Response: Please create a visualization of the ERD diagram for this database. create it as a png that I can download.

Claude: Ran 4 commands, edited 2 files, viewed a file

Your ERD is ready — 5,709 × 4,509 px at 150 DPI, so it's crisp at any zoom level.

The diagram includes all 18 entities laid out across 7 color-coded domains:

- **Blue (dark)** — Member Domain (MEMBER_DIM, MEMBER_ENROLLMENT, EMPLOYER)
- **Blue (medium)** — Plan/Formulary Domain (HEALTH_PLAN, FORMULARY, FORMULARY_DRUG, FORMULARY_TIER)
- **Green** — Accumulator Domain (ACCUMULATOR, ACCUMULATOR_CONTRIBUTION)
- **Brown/Red** — Claims Domain (CLAIM, CLAIM_COST, CLAIM_ADJUSTMENT)

- **Purple** — Drug Reference Domain (DRUG_REFERENCE, PDC_THERAPEUTIC_GROUP, PDC_NDC_GROUP_MAP)
- **Gold** — Provider Domain (PRESCRIBER, PHARMACY)
- **Dark Red** — Authorization Domain (PRIOR_AUTH)

Each entity box shows PK fields (yellow highlight), FK fields (blue highlight), and attributes, with directional arrows and cardinality labels (1:1, 1:M) on all relationships.

File: **Pharma claims ERD.png**

Response: Can you summarize our discussion about the database requirements and design considerations?

Claude: Here's a summary of our database design session:

Purpose & Scope

You're building a post-adjudication analytics database for a health insurance company focused on pharmaceutical claims. The primary use cases are drug cost decomposition and medication adherence measurement using Proportion of Days Covered (PDC).

Core Design Decisions

Member Management

- Members are the central entity, tracked with full SCD Type 2 versioning — every change to demographics, marital status, employer, health plan, etc. creates a new versioned row
- A constant MEMBER_ID spans all versions; a system-generated surrogate key (MEMBER_SK) uniquely identifies each version
- Claims point to MEMBER_SK so you always know which member version was active at fill date

Claims

- Full adjustment and reversal history is preserved — every version of a claim is kept, with IS_CURRENT_VERSION flagging the final adjudicated state
- Cost is decomposed into plan paid, member copay, member coinsurance, member deductible, and other payer amounts
- Key claim attributes include fill date, days supply, quantity dispensed, quantity prescribed, refill number, and new vs. refill flag

Drug Reference

- NDC-to-GPI-14 mapping is maintained internally (not from a vendor)

- The full GPI-14 hierarchy (GPI-2 through GPI-14) is stored explicitly for query performance
- Package size and unit dose information is included to support days supply validation

PDC & Adherence

- PDC is measured at the NDC level but with therapeutic grouping — if a member switches NDCs within the same drug class (e.g., one statin to another), those claims are grouped together for adherence measurement
- A PDC_THERAPEUTIC_GROUP table defines cohorts, and PDC_NDC_GROUP_MAP maps NDCs to those cohorts with effective dates

Formulary

- The formulary is fully versioned with effective and expiration dates at both the drug-tier assignment level and the tier cost sharing rules level
- Tier rules capture copays (retail 30/90, mail 90), coinsurance percentages and min/max, and whether cost sharing is subject to the deductible
- Prior authorizations and formulary exceptions are tracked through their full lifecycle: request date, approval/denial, approving clinician, effective dates, appeal outcome, and linkage to the specific drug and claim

Accumulators

- Individual and family deductible and out-of-pocket accumulators are tracked per benefit period
- Cross-accumulator relationships are supported — when an individual's claim spend contributes to both the individual and family bucket, both contributions are recorded and linked

Providers

- Prescribers are captured with NPI, name, specialty, taxonomy code, and geographic demographics
- Pharmacies are captured with NPI, name, type (retail, mail order, specialty, compounding, long-term care), and location — pharmacy type matters because it affects days supply logic in PDC calculations

Final Schema

The design landed on 18 entities across 7 domains: Member, Plan/Formulary, Accumulators, Claims, Drug Reference, Provider, and Authorization.

Response: Please create a comprehensive README that explains all entities, relationships, and design considerations. Include rationales for key decisions we made during our discussion.

Claude: Here's your comprehensive README. Here's what's covered across its 9 sections:

Section 1 — Project Overview defines the purpose (post-adjudication analytics, not real-time processing) and lays out the six primary use cases as a quick-reference table.

Section 2 — Design Philosophy articulates the three foundational principles we established early: analytics-first architecture, member-centricity, and the deliberate exclusion of DUR/clinical edit data.

Section 3 — Schema Overview gives a quick visual inventory of all 18 entities across 7 domains.

Section 4 — Domain Reference is the bulk of the document — every entity gets a full column table with types and notes, plus narrative explanation and rationale for non-obvious design choices (SCD2 mechanics, NDC normalization, cross-accumulator pattern, etc.).

Section 5 — Entity Relationship Summary is a clean table of all 23 relationships with cardinality and plain-English descriptions.

Section 6 — Analytics Patterns documents the six key query patterns (final claim filter, PDC calculation steps, point-in-time member lookup, formulary compliance, accumulator balance, adjustment history) with SQL examples.

Section 7 — Implementation Notes covers indexing recommendations, SCD2 maintenance procedures, claim adjustment processing steps, and NDC format normalization rules.

Section 8 — Design Decisions & Rationale is a consolidated table of every major architectural choice we made during our discussion — with the reasoning behind each one. This is the section that preserves the "why" for future developers.

Section 9 — Glossary defines 20 domain terms (AWP, GPI, NDC, PDC, SCD2, etc.) for anyone new to the healthcare claims space.

File: pharma_claims_README.pdf